

yhlleo/DeepSegmentor: A Pytorch implementation of DeepCrack and RoadNet projects. · GitHub

 github.com/yhlleo/DeepSegmentor

yhlleo

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A Pytorch implementation of DeepCrack and RoadNet projects.

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Contributors

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Issues

 306
Stars

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Pytorch 0.4.1

last commit december 2024

Maintained? yes

contributions welcome

 Open in Colab

DeepSegmentor

A Pytorch implementation of DeepCrack and RoadNet projects.

1.Datasets

- [Crack Detection Dataset](#)
- [Multi-task Road Detection Dataset](#)

Please download the corresponding dataset and prepare it by following the [guidance](#).

2.Installation

We provide an user-friendly configuring method via [Conda](#) system, and you can create a new Conda environment using the command:

```
conda env create -f environment.yml
```

3.Balancing Weights

We follow the [Median Frequency Balancing](#) method, using the command:

```
python3 ./tools/calculate_weights.py --data_path <path_to_segmentation>
```

4.Training

Before the training, please download the dataset and copy it into the folder datasets.

Crack Detection

```
sh ./scripts/train_deepcrack.sh <gpu_id>
```

Road Detection

```
sh ./scripts/train_roadnet.sh <gpu_id>
```

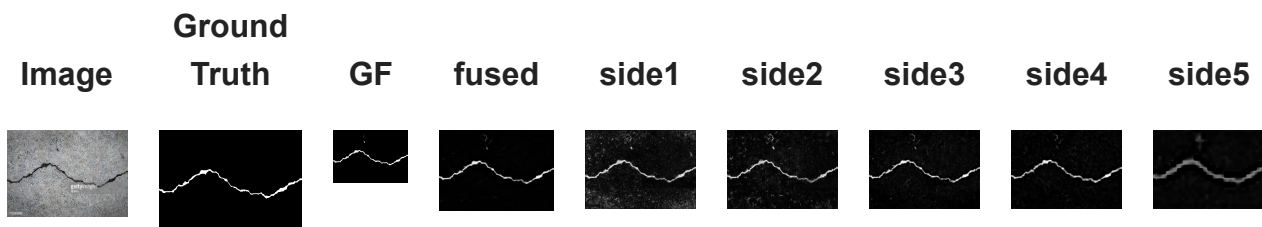
We provide our pretrained models here:

Model	Google Drive	Baidu Yun	Others
DeepCrack	 [link]	 [link] (psw: 3fai)	Fine-tuned
RoadNet	 [link]	 [link] (psw: c2gi)	Roughly trained
RoadNet++	[link]	[link]	-

5.Testing

Crack Detection

```
sh ./scripts/test_deepcrack.sh <gpu_id>
```

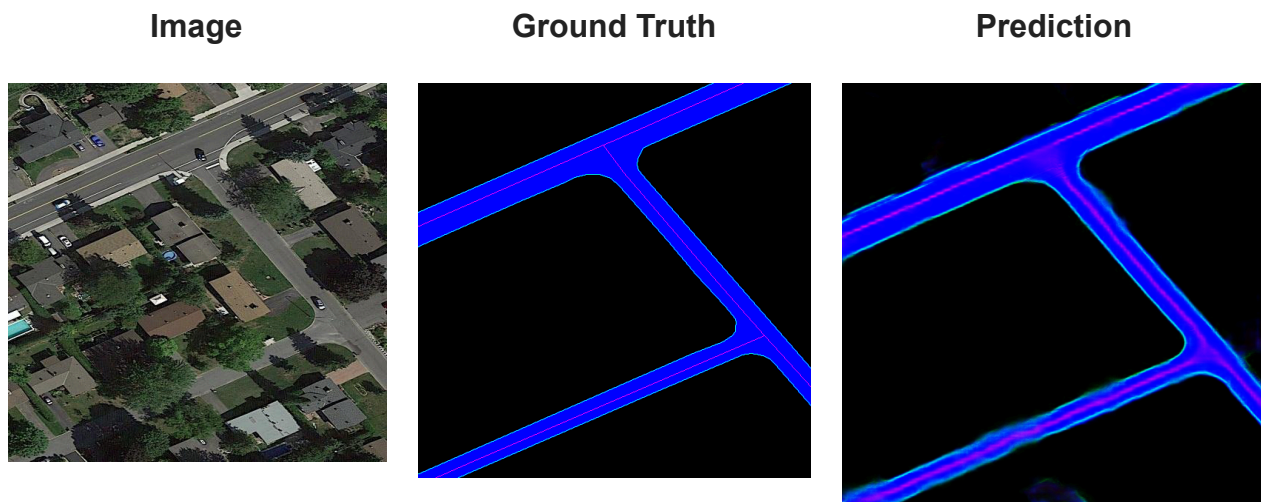


[\[See more examples >>>\]](#)

PS: Once you downloaded the pretrained models, you can follow the test example in the [jupyter notebook](#) to test your own images.

Road Detection

```
sh ./scripts/test_roadnet.sh <gpu_id>
```



[\[See more examples >>>\]](#)

6.Evaluation

Metrics (appeared in our papers):

Metric	Description	Usage
P	Precision, $TP / (TP + FP)$	segmentation
R	Recall, $TP / (TP + FN)$	segmentation
F	F-score, $2PR / (P + R)$	segmentation
TPR	True Positive Rate, $TP / (TP + FN)$	segmentation
FPR	False Positive Rate, $FP / (FP + TN)$	segmentation
AUC	The Area Under the ROC Curve	segmentation
G	Global accuracy, measures the percentage of the pixels correctly predicted	segmentation
C	Class average accuracy, means the predictive accuracy over all classes	segmentation
I/U	Mean intersection over union	segmentation
ODS	the best F-measure on the dataset for a fixed scale	edge, centerline
OIS	the aggregate F-measure on the dataset for the best scale in each image	edge, centerline

Metric	Description	Usage
AP	the average precision on the full recall range	edge,centerline

Note: If you want to apply the standard non-maximum suppression (NMS) for edge/centerline thinning. Please see more details in [Piotr's Structured Forest matlab toolbox](#) or some helper functions provided in the [hed/eval](#).

[\[See more details \(Evaluation + Guided Filter + CRF\) >>>\]](#)

Usage:

```
cd eval
python eval.py --metric_mode prf --model_name deepcrack --output deepcrack.prf
```

[\[Display the accuracy curves >>>\]](#)

Acknowledgment

- This code is based on the [pytorch-CycleGAN-and-pix2pix](#). Thanks to the contributors of this project.
- If you are familar to [Google Colab](#), there is an implementation of [Colab version](#) (provided by [DZDL/DeepSegmentor](#)). Besides, there is a WebApp version of this project: [crack-detector](#).

References

If you take use of our datasets or code, please cite our papers:

```
@article{liu2019deepcrack,
  title={DeepCrack: A Deep Hierarchical Feature Learning Architecture for Crack Segmentation},
  author={Liu, Yahui and Yao, Jian and Lu, Xiaohu and Xie, Renping and Li, Li},
  journal={Neurocomputing},
  volume={338},
  pages={139--153},
  year={2019},
  doi={10.1016/j.neucom.2019.01.036}
}
```

```
@article{liu2019roadnet,
  title={RoadNet: Learning to Comprehensively Analyze Road Networks in Complex Urban Scenes from High-Resolution Remotely Sensed Images},
  author={Liu, Yahui and Yao, Jian and Lu, Xiaohu and Xia, Menghan and Wang, Xingbo and Liu, Yuan},
  journal={IEEE Transactions on Geoscience and Remote Sensing},
  volume={57},
  number={4},
```

```
pages={2043--2056},  
year={2019},  
doi={10.1109/TGRS.2018.2870871}  
}
```

If you have any questions, please contact me without hesitation (yahui.cvr AT gmail.com).